

Flexible machine-learning-based imputation with options for sequential imputation and predictive mean matching. `vimpute()`

Eileen Vattheuer

Eileen.Vattheuer@statistik.gv.at

Nina Niederhametner

Nina.Niederhametner@statistik.gv.at

Alexander Kowarik

Alexander.Kowarik@statistik.gv.at

Johannes Gussenbauer

Johannes.Gussenbauer@statistik.gv.at

Q2026 - ŠIBENIK, CROATIA, June 2026

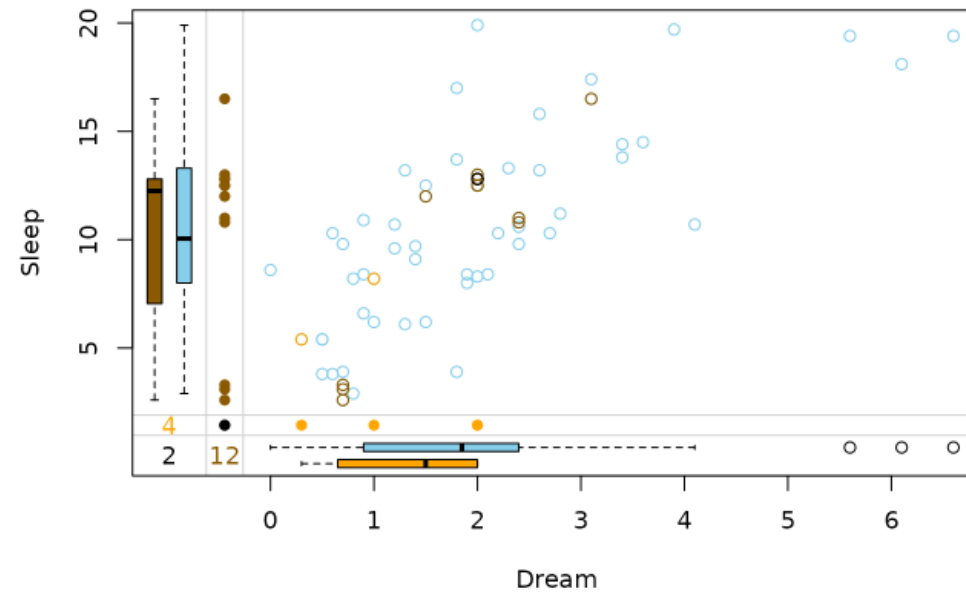
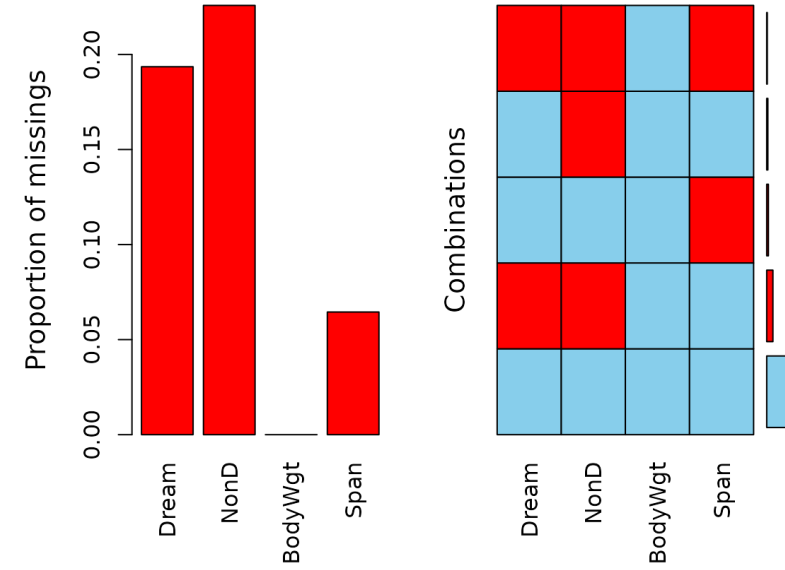
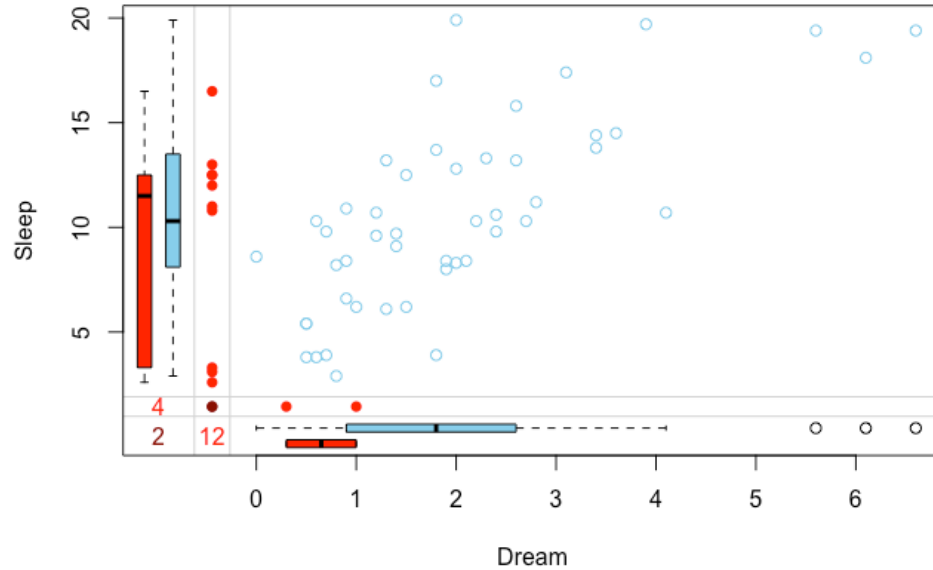
www.statistik.at

The R package **VIM**

<https://github.com/statistikat/VIM/>

<https://cran.r-project.org/web/packages/VIM/index.html>

Visualization of Missing Values



Imputation Methods



Donor based methods

```
kNN(data, variable = ..., k = 5, dist_var = ..., ...)
```

```
hotdeck(data, variable = ..., ord_var = ..., domain_var = ..., ...)
```

```
matchImpute(data, variable = ..., match_var = ..., ...)
```

Model based methods

```
regressionImp(formula = ..., data = ...family = ...,
```

```
# Iterative robust model-based imputation
```

```
irmi(x = data, maxit = 100, noise = ..., robust = ..., ...)
```


Imputation Methods



ML based methods

```
rangerImpute(formula = ..., data = ..., ...)
```

```
xgboostImpute(formula = ..., data = ..., ...)
```

```
transformerImpute(data = ..., target = ..., cat_vars = NULL, ...)
```

Key Features and Advantages of `vimpute()`

- Unified framework
- Single function to handle imputation of multiple variables
- Variable-specific models
- PMM
- Automatic iteration & convergence check
- Supports modern ML models via **mlr3**
- Hyperparameter tuning possible
→ finds best model settings automatically

Age	Education	Yearly Income
30	Post-Secondary	NA
NA	Secondary	28000
52	Tertiary	NA
18	NA	0

↓
robust
regression

↓
random forest
+
tuning

↓
xgboost
+
PMM

Workflow

Check data

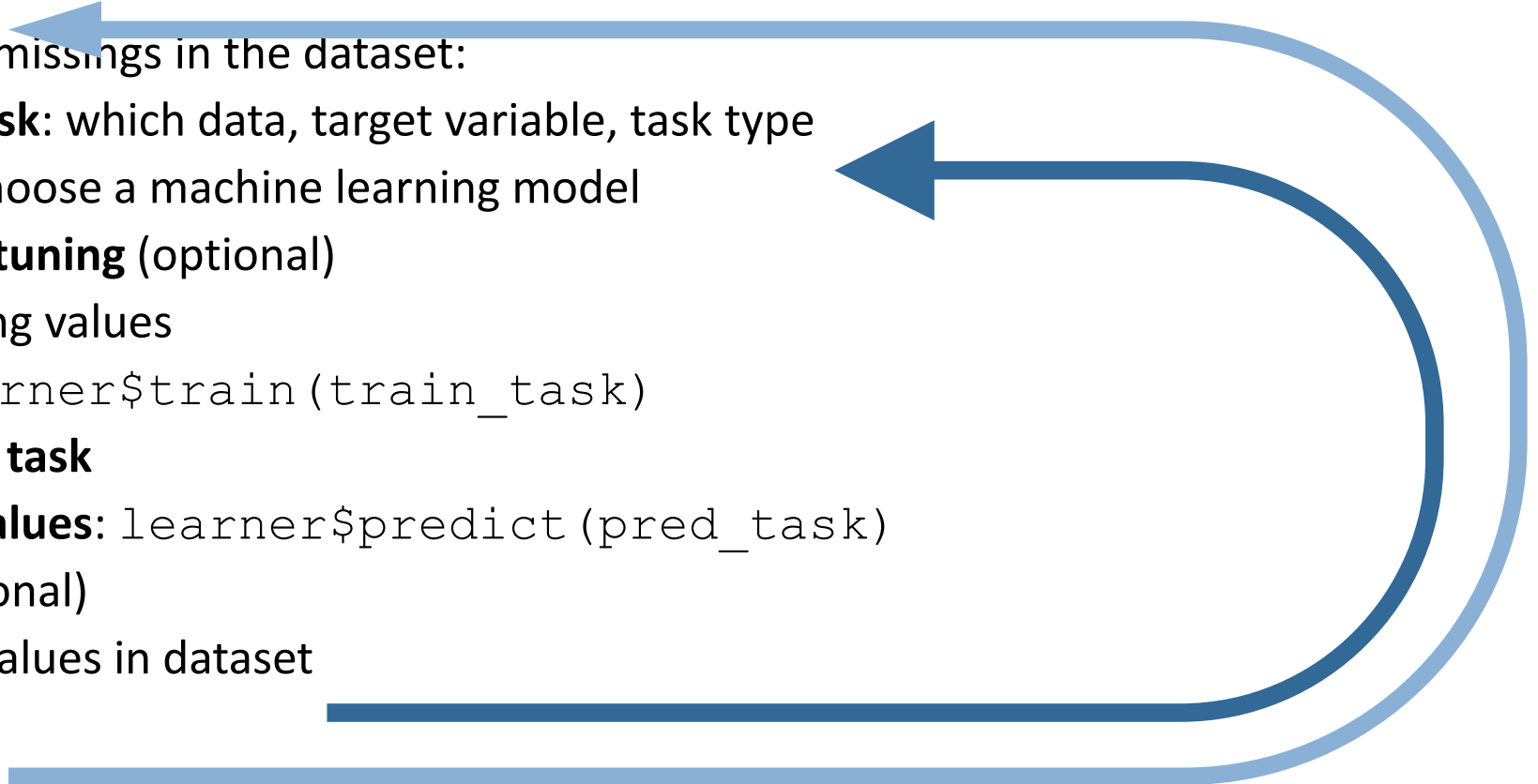
For n iterations:

For each variable with missings in the dataset:

- **Create training task**: which data, target variable, task type
- **Create learner**: choose a machine learning model
- **Hyperparameter tuning** (optional)
- Handling of missing values
- **Train model**: `learner$train(train_task)`
- **Create prediction task**
- **Predict missing values**: `learner$predict(pred_task)`
- Apply **PMM** (optional)
- **Replace** missing values in dataset

Check stopping criteria

Return final dataset



Subset of the available parameters

```
vimpute(  
  data = as.data.table(sleep),  
  considered_variables = c("BodyWgt", "BrainWgt", "Dream"),  
  method = list(BodyWgt = "ranger",  
                 BrainWgt = "xgboost",  
                 Dream = "robust"),  
  pmm = list(BodyWgt = TRUE,  
              BrainWgt = FALSE,  
              Dream = FALSE),  
  formula = FALSE,  
  nseq = 10,  
  tune = FALSE )
```


mlr3 Integration

- Modular, principled framework: Task, Learner, Tuning
- Unified interface: one consistent interface for all models for different ML models
- Large & extensible ecosystem
- Integrated hyperparameter tuning
- Parallelization support

→ **Structured, flexible, modern ML framework with tuning and parallelization built-in**

Methods

ranger – random forest 

xgboost – extreme gradient boosting 

regularized – regularized lm 

robust – robust linear regression 

- Not directly from mlr3
- Wraps `robustbase::lmrob()` in an mlr3-compatible `LearnerRegr` object
- Inspired by `irmi`

Multiple imputation / Model uncertainty

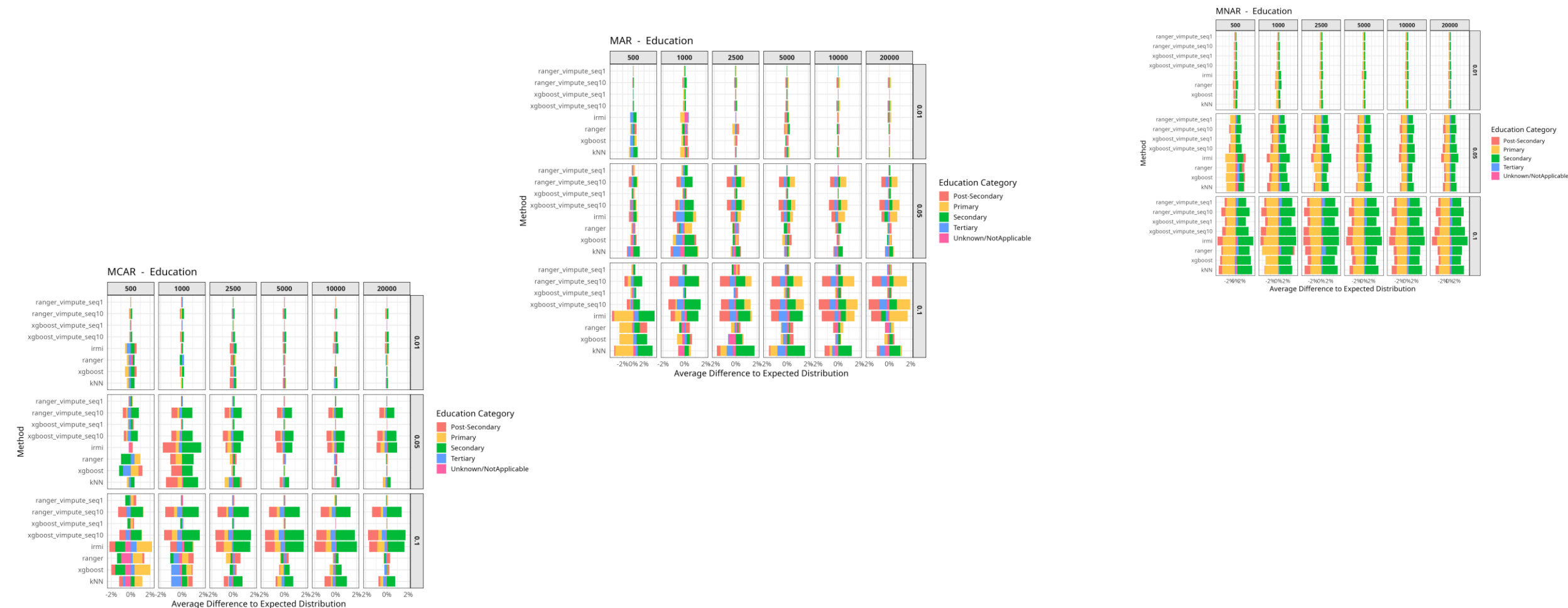


(Robust) bootstrap resampling available to account for model uncertainty



Different strategies to generate multiple imputed values available

Simulation study -> work in progress, publication will follow



Conclusion

Our approach to bring everything into one unified framework

Expandable framework

- Options for future extensions
 - Easy to integrate
-

Todos: Trade off between prediction performance and to preserve the dataset's distributions

- Stopping criteria also based on distribution
-